

Hey AI, Make This Vibrate! : A Demonstration of Rapid Haptic Prototyping Using a Generative Text-to-Vibration Model

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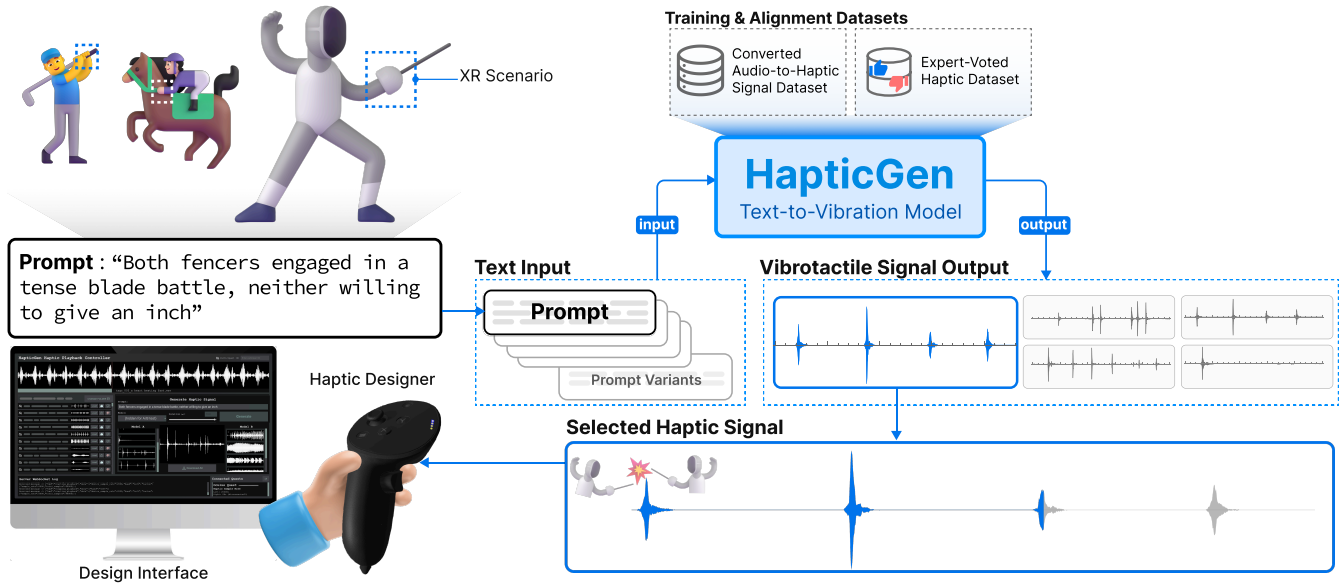


Figure 1: Overview of HapticGen. Designers input a text prompt, and the system generates multiple haptic signals played on VR controllers via a desktop interface.

CCS Concepts

• **Human-centered computing** → Haptic devices; Virtual reality; • **Hardware** → Tactile and hand-based interfaces; • **Computing methodologies** → Model development and analysis.

*Both authors contributed equally to the paper.

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Keywords

Haptics, Designers, Generative AI, Extended Reality

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1 Introduction

Vibrotactile feedback enriches user experience in XR, gaming, and assistive technologies, but designing such feedback remains difficult due to the lack of accessible authoring tools. Prior methods often rely on waveform editors or audio-to-haptic conversion, both

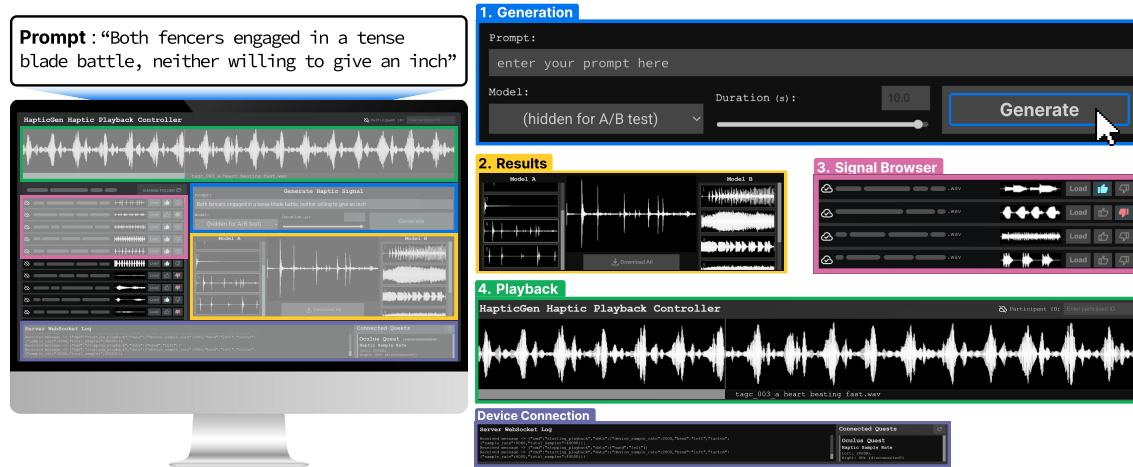


Figure 2: Our HapticGen interface is divided into four numbered sections: Generation, Results, Signal Browser, and Playback pane.

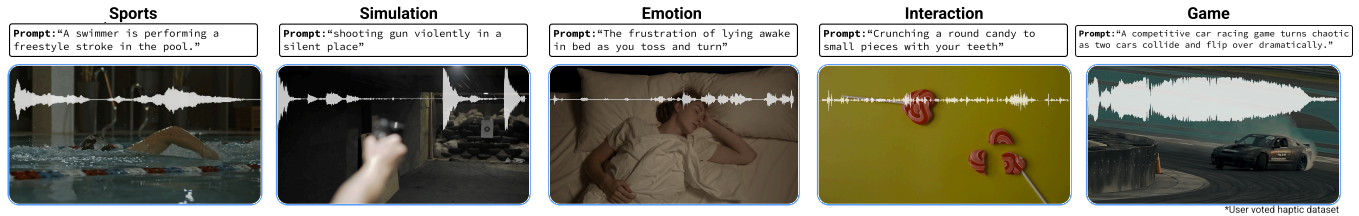


Figure 3: Example prompts provided by participants, along with the corresponding vibrations generated by HapticGen. Participants voted on the vibration outputs after experiencing the haptic feedback on Quest controllers.

requiring significant expertise. We propose **HapticGen**, the first generative model that produces haptic signals directly from text prompts, reducing cognitive and technical barriers for designers[3].

2 System Overview

HapticGen builds on a MusicGen-style transformer trained on 335k converted audio-haptic samples and further fine-tuned on expert-rated prompt-vibration pairs. The interface allows users to input a textual scenario, view prompt variations generated via LLMs, and preview the output vibrations on connected Meta Quest controllers[1, 2] without wearing a HMD (head-mounted display). Users can vote signals up/down, save preferred outputs, and iterate via prompt editing.

3 Demonstration Description

In this hands-on demo, attendees can:

- Input a descriptive text prompt (e.g., *"a distant thunderstorm rumbling"*)
- Generate and preview five vibration variants on Quest Pro controllers
- Rate each signal's match and expressiveness using thumbs up/down
- Modify the prompt and observe how subtle changes affect tactile output

All interactions occur on a laptop interface, with no need for a VR headset, and the setup is fully standalone.

4 Impact and Future Work

HapticGen makes haptic design more accessible to artists, designers, and researchers without deep technical skills. We believe this framework opens new possibilities for expressive and rapid prototyping in haptics, and we aim to extend it to multi-modal and personalized feedback.

5 Project Web

The model and datasets are open source:

- <https://hapticgen.hcitech.org/>
- <https://github.com/HapticGen/HapticGen>

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